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Von Willebrand factor A1 blockade prevents platelet-mediated sustained occlusion for the treatment of arterial thrombosis



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Arterial thrombosis is a leading cause of death and disability worldwide. Administration of tissue plasminogen activator (tPA) remains the current noninvasive clinical gold standard but is ineffective for many patients. Von Willebrand factor (VWF) is mechanistically critical in arterial thrombosis and has been studied as an alternative target for thrombolytic therapy. This study utilizes a microfluidic system of arterial thrombosis to evaluate VWF-A1 domain inhibitor aptamer (BB-031), aiming to understand VWF-mediated recanalization (vessel reopening) and compare efficacy of BB-031 to Alteplase and Tenecteplase. Thrombotic occlusion is simulated in a microfluidic device, and thrombi are allowed to retract and remodel for up to 6 hr. During a subsequent 2 h treatment reperfusion period (not disruptive to the original thrombus), thrombus morphology, composition, and channel patency (openness) are analyzed. Thrombi substantially remodel post-occlusion. BB-031 treatment results in an inability to maintain thrombus and significantly improves microfluidic patency compared to vehicle and tPA. Acute ischemic stroke patient samples demonstrate improved microfluidic patency with BB-031 compared to vehicle. Here we establish an in vitro/ex vivo platform to study arterial occlusion, clot retraction, drug delivery, and recanalization/patency, and implement this platform to demonstrate BB-031 could be a safe and efficacious therapeutic alternative to tPA.

Arterial thrombosis is a major global health burden, contributing to conditions such as acute coronary syndrome (ACS) and acute ischemic stroke (AIS)¹. The World Health Organization estimates that 17.9 million people die from cardiovascular disease annually with roughly 85% of them being due to heart attack or stroke². The World Stroke Organization estimates that over 12 million new strokes occur annually, with nearly half of survivors experiencing significant morbidity, including paralysis, speech impairment, and cognitive deficits^{3,4}. In addition to its profound human toll, arterial thrombosis imposes substantial economic costs, with annual healthcare expenses in the United States alone exceeding \$150 billion for ACS and \$40 billion for stroke^{5,6}.

Despite advancements in management, therapeutic options towards arterial thrombolysis remain limited. Currently, the only FDA-approved pharmacological interventions for thrombolysis in ACS and AIS are tissue

plasminogen activator (tPA) formulations Tenecteplase and Alteplase⁷. These agents act by converting plasminogen to plasmin to potentiate fibrinolysis and thrombus dissolution⁷. However, the efficacy of tPA is constrained by several critical limitations. Administration of tPA in patients with ST-elevation myocardial infarctions who are unable to undergo percutaneous intervention is limited to 12 h following symptom onset⁸. Successful recanalization is achieved in only about 30% of AIS cases, and its therapeutic window is narrow: less than 4.5 h from symptom onset⁷. This short timeframe excludes many patients, and particularly those in rural or resource-constrained settings where delays in diagnosis and treatment are common⁹. Additionally, irreversible activation of plasmin by tPA significantly increases the risk of intracranial hemorrhage and systemic bleeding^{10,11}. Tenecteplase offers enhanced fibrin specificity and a longer half-life, though it has not demonstrated meaningful improvements in

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efficacy or safety outcomes^{12,13}. Endovascular thrombectomy has shown safety and efficacy for treatment of AIS caused by large vessel occlusion with a therapeutic window of up to 24 h^{14,15}. However, it remains inaccessible to many patients due to its reliance on specialized expertise and infrastructure, as well as patient-specific considerations, and is an invasive procedure. A more robust and efficacious thrombolytic therapy would be transformative for patient care and outcomes.

The molecular composition of thrombi adds a layer of complexity to management of arterial thrombosis. Evaluating human thrombus composition *in vivo* presents substantial logistical challenges, including uncertainty in thrombus maturation age and morphology alterations introduced by manipulation, underscoring a need for *in vitro* and *ex vivo* models that can accurately recapitulate thrombus features to better investigate therapeutic efficacy and mechanism. Studies have shown that arterial thrombi, and specifically AIS thrombi, are often heterogeneous, comprising plasma-derived proteins and cells in varying proportions^{16–18}. Notably, these thrombi are frequently rich in platelets and von Willebrand factor (VWF), reflecting the high-shear flow conditions under which these components are incorporated, and contain an outer fibrin shell or a fibrin presence that is twisted with large pores^{16,18–22}. Fibrinolytic therapies primarily target fibrin; however, the unique composition and mechanodynamic characteristics of platelet- and VWF-rich thrombi significantly reduce their susceptibility to these treatments^{18,23}. These challenges have spurred investigations into alternative therapeutic targets, including VWF^{24,25}. Despite promising pre-clinical studies, no alternative therapies have yet demonstrated clinical efficacy, and bleeding risks remain a significant concern.

VWF is a multimeric glycoprotein critical for mediating platelet adhesion and aggregation under both normal and pathological conditions. Following vascular injury, VWF binds to subendothelial collagen and captures platelets via interactions between the VWF-A1 domain and platelet glycoprotein IIb (GPIIb)²⁶. However, under pathological conditions of high shear stress, such as in stenosed or diseased arteries, VWF elongates and extends, exposing its A1 domain and promoting platelet binding^{27,28}, a process required for the formation of platelet-rich thrombi. Furthermore, recent work has demonstrated that VWF can organize into cable-like structures capable of arresting arterial flow during platelet-mediated thrombosis independent of coagulation, highlighting key contributions of VWF to structural integrity and flow-obstructive properties of occlusive thrombi²⁹. Collectively, VWF is thus a compelling target for therapeutic intervention in arterial thrombosis.

Aptamer inhibitors offer many advantages including the ability to modify their half-life, unlimited shelf life, low immunogenicity, and activity that can be readily reversed. Preclinical studies have demonstrated that blocking the VWF-A1 interaction using the aptamer inhibitor BB-031 can restore blood flow in *in vitro* and *in vivo* models of thrombosis, and functional efficacy was highly sensitive to shear-induced exposure of the A1 domain^{30,31}. BB-031 has also shown promise in a Phase I clinical trial (NCT05005520), exhibiting a favorable safety profile, and Phase II is ongoing (NCT06226805). However, the mechanisms underlying VWF-targeted recanalization (blood vessel reopening) remain poorly understood, and its efficacy compared to fibrinolytic agents has yet to be comprehensively evaluated.

The flow-dependent nature of VWF-mediated thrombus formation presents unique challenges for studying recanalization under pathophysiologically-relevant conditions. Existing experimental models often fail to replicate the hemodynamic and temporal remodeling processes characteristic of arterial thrombosis and prolonged thrombotic occlusion. Microfluidic platforms offer a promising solution, allowing customizable channel dimensions, biological surface coating, and fluid dynamic properties. This study thus aimed to develop a microfluidic system of arterial, platelet-rich, occlusive thrombosis to: (i) test therapeutic agents under pathophysiologically-relevant conditions post-occlusion in the setting of thrombus retraction and remodeling, (ii) elucidate mechanisms underlying VWF-targeting therapies, and (iii) compare VWF-targeting therapies to

current fibrinolytic agents *in vitro* and *ex vivo*. We hypothesize that the VWF-targeted therapy (BB-031) evaluated in our arterial thrombosis microfluidic platform will recapitulate recanalization seen previously *in vivo*, and will be more effective at achieving microfluidic patency (openness) compared to Alteplase and Tenecteplase. Herein, our platform suggests VWF-mediated platelet recruitment and reinforcement is a driver of thrombus maintenance and lytic-resistance. We show that inhibition of VWF-A1 with aptamer BB-031 disrupted this dynamic arterial thrombus reinforcement to restore blood flow more effectively than fibrinolytic agents.

Results

Microfluidic occlusive, platelet-rich thrombosis

Occlusive thrombi formed from healthy volunteer whole blood in the stenotic region of the microfluidic device contained high platelet content (Supplementary Movie 1: ‘Thrombus formation.avi’). Representative microfluidic schematic and occlusive clot formation images are shown in Fig. 1A–C. Normalized mean fluorescence intensity (MFI) of a growing thrombus is shown in Fig. 1D with clear lag, growth, and plateau phases of clot formation. Mass outflow measured during clot formation is shown in Fig. 1E. A plateau of mass outflow signifies occlusion in the microfluidic device against an arterial pressure head. A representative z-stack confocal image of thrombus 3 h post-occlusion is shown in Fig. 1F, depicting key compositional features of platelets, VWF, and a fibrin shell. Supplementary Fig. 4A demonstrates that the microfluidic secondary outlet can be utilized to reperfuse the system immediately post-occlusion without disruption of the formed thrombus.

Structural and compositional thrombus changes

In the post-occlusion period in the microfluidic device, thrombus remodeling is apparent with clusters of deposited platelets becoming globular and denser (Fig. 2A). Quantified thrombus pore area (Fig. 2B–C) demonstrates significant reduction in pore area 3 and 6 h post-occlusion, while pore area from the end of clot formation (occlusion) to 0 h post-occlusion demonstrated no significant change in thrombus pore area. Timing between occlusion and 0 h post-occlusion images reflect the time required to set up reperfusion in the microfluidic platform.

Confocal microscopy z-stack images of the thrombus structure show the presence of a fibrin shell (Fig. 2D), which was apparent 0 h post-occlusion and remained intact 3 and 6 h post-occlusion. Thrombi also contained platelets and VWF in the thrombus core. Clot remodeled 6 h post-occlusion appeared to increase platelet and VWF density in the thrombus core, which remained surrounded by fibrin fibers.

VWF involvement in the developing thrombus and contribution of newly arriving platelets to thrombus structure

Confocal microscopy images of platelet aggregates that grew during thrombus formation in the microfluidic device appear to also contain VWF strands that cloak platelet clusters (Fig. 3A). As a separate, differentially labeled population of platelets from the same parent blood sample was introduced during thrombus formation in the microfluidic device (CD41_{Rep}) (Fig. 3B), VWF appears to bridge the newly captured platelets to continuously grow the platelet aggregate at high shear. These newly reperfused platelets appear to contain clusters of granular VWF distribution suggesting that these are naïve platelets that have not undergone degranulation and activation.

Dynamic platelet contributions during thrombus retraction were observed. As depicted in Fig. 3C, the platelet population involved in initial thrombus formation (CD41_{Throm}) retracts toward the channel center, demonstrated by 2D surface area maps. Quantification of CD41_{Throm} surface area was performed in thrombi exposed to newly reperfused platelets and in thrombi maintained under static conditions at 6 h post-occlusion. The data revealed a 6.9% reduction in thrombus surface area 2 h after the introduction of new platelets, compared to a 1.6% reduction in thrombi under similar conditions without the perfusion of new platelets (Fig. 3D).

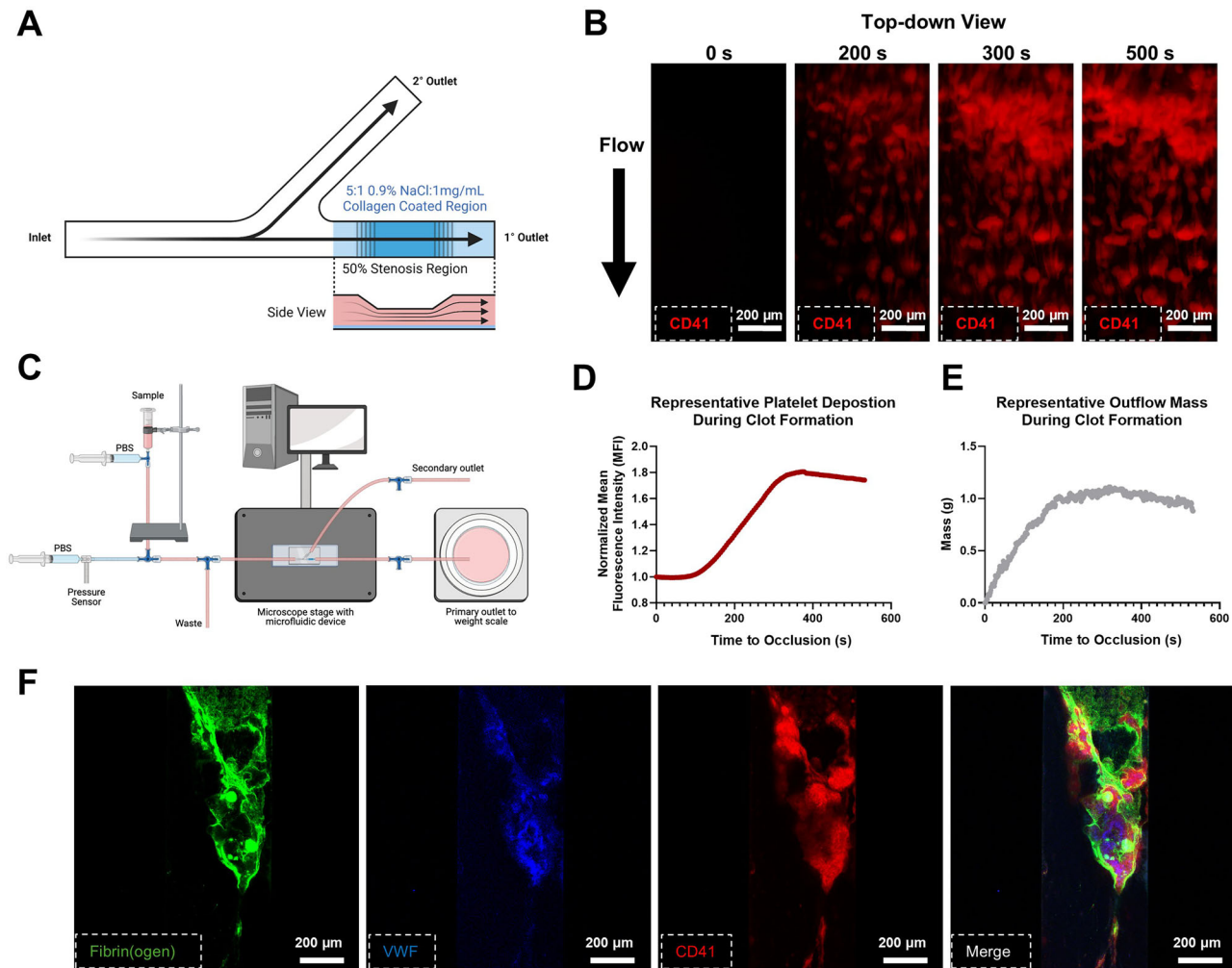


Fig. 1 | Stenotic microfluidic model and kinetic monitoring of occlusive thrombus at arterial flow. **A** Schematic representation of the microfluidic flow chamber and **(B)** Representative fluorescence microscopy images (Zeiss, Axio Observer 7) of forming thrombus to occlusion. **C** Schematic representation of experimental setup with constant pressure blood flow, fluorescent microscopy, and outflow mass measurement. **D** Representative platelet deposition quantification and

(E) outflow mass over time to occlusion. **F** Representative z-stack image (a maximum intensity projection from 21 slices with 4.2 μm step size) captured using confocal microscopy (Olympus FV4000) of thrombus structure 3 h post-occlusion. Created in BioRender. Kar, R. (2026) <https://BioRender.com/7lg93qj>.

VWF-A1 inhibition reduces new platelet deposition following thrombosis and retraction

Microfluidic experiments revealed that continual delivery of blood to an occlusive thrombus (containing $\text{CD41}_{\text{Throm}}$) 3 h post-occlusion led to substantial recruitment of new platelets (CD41_{Rep}) and VWF, which collectively contributed to sustained channel occlusion (Fig. 4A). CD41_{Rep} was observed predominantly at the interface between $\text{CD41}_{\text{Throm}}$ and incoming blood flow, with VWF fibers clearly bridging these platelet populations. Blood treated with the VWF-A1 aptamer BB-031 reduced the accumulation of new platelets, as evidenced by reduction in CD41_{Rep} . Quantitative analysis demonstrated a marked reduction in the integrated density of CD41_{Rep} in BB-031-treated thrombi compared to vehicle-treated controls after 2 h reperfusion ($p = 0.0172$) (Fig. 4B). A depiction of maintenance of thrombus and inhibition of continual platelet recruitment mediated by VWF-A1 is presented in Fig. 4C.

BB-031 achieves patency more effectively than tPA

Platelet count and hematocrit from healthy volunteer donors showed no significant differences between donations when compared among vehicle, BB-031, Alteplase, and Tenecteplase treatment comparisons (Supplementary Table 1). Post-occlusion (3 or 6 h), VWF-A1-binding aptamer BB-031 increased mean outflow mass (indicating recanalization and patency) more

than vehicle, Alteplase, and Tenecteplase over the 2 h treatment period (Fig. 5A–B). However, 0 h post-occlusion minimal mass outflow was observed (Supplementary Fig. 4B). 3 h post-occlusion, BB-031 mass outflow was 0.951 ± 1.279 g (mean $\pm$ standard deviation) compared to 0.156 ± 0.266 g with vehicle, 0.487 ± 0.597 g with Alteplase, and 0.249 ± 0.329 g with Tenecteplase. 6 h post-occlusion, BB-031 increased mass outflow by 1.354 ± 1.433 g compared to 0.297 ± 0.497 g with vehicle, 0.482 ± 0.725 g with Alteplase, and 0.326 ± 0.373 g with Tenecteplase. Inclusive of both timepoints, BB-031 maximum outflow mass (i.e., bulk flow) was significantly higher compared to controls and other treatment groups (Fig. 5C). Representative movie files depict the 2 h reperfusion period for vehicle treatment experiments 3 h post-occlusion (Supplementary. Movie 2: ‘3 h post occlusion vehicle reperfusion.avi’) and 6 h post-occlusion (Supplementary. Movie 3: ‘6 h post occlusion vehicle reperfusion.avi’).

AIS patient samples achieve patency with VWF-A1 inhibition

AIS patient characteristics, including sex, age, antiplatelet medications, and select complete blood count parameters are shown in Supplementary Table 2. In microfluidic testing with AIS patient samples, during the 2 h treatment reperfusion period, outflow mass was negligible with vehicle treatment but substantially higher in the BB-031 treated samples (Fig. 5D).

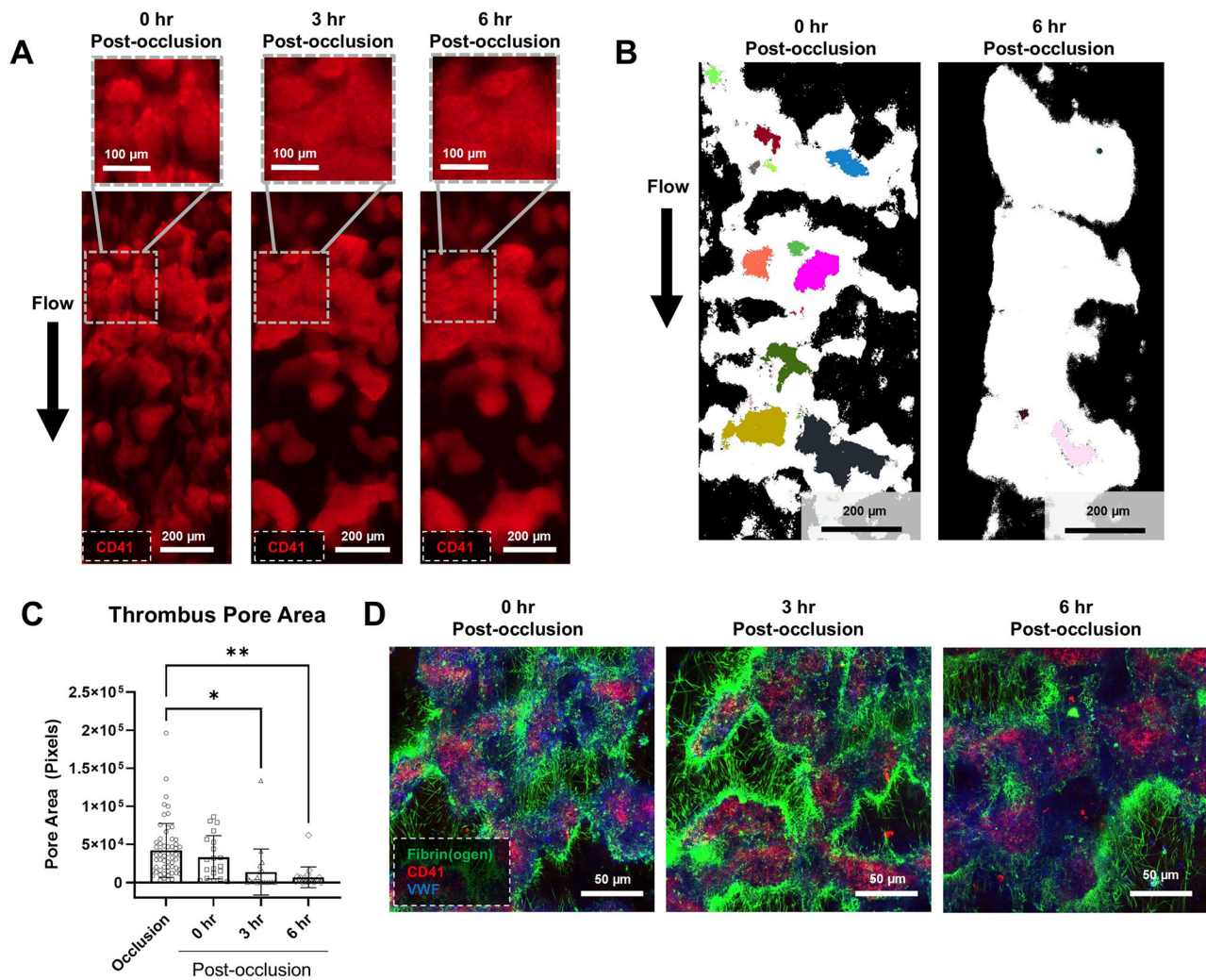


Fig. 2 | Dynamic alteration to thrombus morphology post-occlusion.

A Representative fluorescence microscopy images (Zeiss, Axio Observer 7) from thrombi (CD41 +), 0, 3, and 6 h post-occlusion, with insets showing a closer look at structural changes over time and pore reduction. **B** Representative binarized pore area visualization from fluorescence microscopy images 0 and 6 h post-occlusion. **C** Corresponding pore area quantification immediately after thrombus formation (occlusion, $n = 60$ independent experiments) and 0, 3, or 6 h post-occlusion ($n = 20$ /

group) (occlusion versus 3 h pore area $p = 0.0183$; occlusion versus 6 h pore area $p = 0.0048$). **D** Composition of aging thrombus demonstrating dynamic spatio-temporal distribution of fibrin, platelets, and VWF is visualized in z-stack images (a maximum intensity projection from 18 slices with 5 μm step size) captured using confocal microscopy (Olympus FV4000). * $p < 0.05$; ** $p < 0.01$. All error bars represent standard deviation.

Quantified maximum mass outflow demonstrates a significant improvement in channel patency with AIS patient samples when treated in vitro with BB-031 compared to vehicle (Fig. 5E).

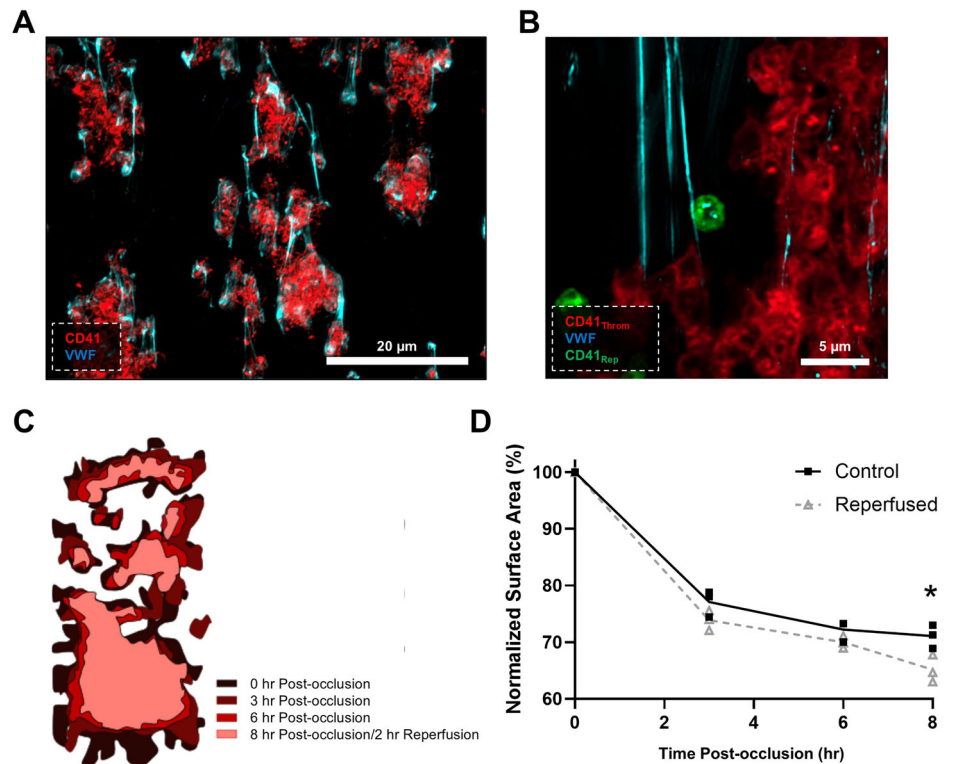
Discussion

This study provides new insights into the mechanisms of thrombus remodeling and VWF-mediated platelet aggregation under high-shear and post-occlusion, demonstrating the potential of targeting this pathway to improve outcomes in patients suffering from sequelae of arterial thrombosis, and suggesting possible mechanism for the previous findings of in vivo recanalization with A1 aptamer inhibition. Using a custom microfluidic platform, we developed an in vitro/ex vivo system capable of simulating occlusive arterial, platelet-rich thrombi, enabling controlled testing of therapeutic interventions. Important for the translation to in vivo efficacy, we observe similar thrombus structures and composition as reports on human AIS thrombi, specifically an outer fibrin shell and high proportion of platelets and VWF¹⁸. Our findings reveal that the VWF-A1 domain plays a critical role in maintaining and sustaining occlusive thrombus by mediating continued platelet recruitment. Similar findings have been reported in murine stroke models using VWF-A1:GPIIb inhibitors³², ADAMTS13^{33,34},

or VWF-targeted plasminogen activation to degrade VWF³⁵, as targeting thrombolytics. However, efficacy with human samples in mature thrombi have not yet been investigated. Importantly, the microfluidic platform utilized here allows for real-time imaging over thrombus maturation stages, and the aptamer BB-031 effectively inhibits this VWF-A1:GPIIb interaction in human samples, reducing thrombus growth and enhancing channel patency more effectively than Alteplase and Tenecteplase, which act as fibrinolytic agents, after delayed treatment. These data suggests that the aptamer BB-031 is acting through its effect on continued platelet recruitment rather than only “dissolving” pre-existing thrombi. As an aptamer-based technology, BB-031 activity can also be readily reversed to reduce bleeding risk following AIS. These results support BB-031 as a promising mechanism-based therapy for arterial thrombosis, targeting pathways unaddressed by current fibrinolytics, and explain why an inhibition strategy can result in recanalization.

Arterial thrombi undergo significant remodeling over time, resulting in reduced porosity and increased density. Our results demonstrated this process kinetically in real-time, with thrombi showing a significant reduction in pore area and increasing density 3 and 6 h post-occlusion. This structural transformation aligns with prior reports attributing thrombus

Fig. 3 | VWF involvement in developing thrombus and contribution of new platelets to thrombus size post-occlusion. **A** Representative z-stack image (a maximum intensity projection from 20 slices with 1.4 μm step size) captured using confocal microscopy (Olympus FV4000) of developing thrombus formed at high shear flow with VWF fibers strewn across deposited platelets. **B** Newly reperused platelets (CD41_{Rep}, green) attach to VWF (cyan) allowing connection to original thrombus platelets (CD41_{Throm}, red). Z-stack image (a maximum intensity projection from 20 slices with 1.4 μm step size) captured using confocal microscopy (Zeiss LSM900) taken ~80 microns above collagen-coated surface. **C** Representative binarized fluorescent microscopy images (Zeiss, Axio Observer 7) depict surface areas of original thrombus over time post-occlusion. **D** Surface area quantification normalized to maximum surface area immediately post-occlusion for thrombus over time (0 h to 6 h) grouped by exposure to newly reperused platelets from 6 h to 8 h (endpoint 8 h control (squares) versus exposure to reperused platelets (triangles) $p = 0.0397$, $n = 3$ independent experiments). * $p < 0.05$.



contraction to platelet cytoskeletal rearrangements, which stabilize the thrombus and reduce its permeability to therapeutic agents^{36,37}. Confocal microscopy revealed that these remodeled thrombi maintained a fibrin shell surrounding a core of densely packed platelets and VWF. This resulting core-shell structure creates additional barriers to fibrinolytic penetration, potentially contributing to the limited efficacy of Alteplase and Tenecteplase. These findings underscore the importance of therapies that can disrupt thrombus progression and overcome the mechanical resistance posed by dense thrombi.

VWF is classically known for its role in hemostasis, bridging sub-endothelial collagen to platelets via its A1 domain binding to platelet GPIIb α during vascular injury. Beyond this function, studies have shown that under high-shear conditions, VWF elongates independently of collagen exposure, enabling platelet recruitment and platelet-platelet bridging in a process termed activation-independent aggregation³⁸. This mechanism is particularly relevant to arterial thrombosis, where high shear stress promotes thrombus growth even in the absence of overwhelming vascular damage. Our study demonstrates this phenomenon: following thrombus occlusion and contraction, residual blood flow carries fresh platelets to the thrombus interface, where VWF bridges these platelets to the existing thrombus, initiating a self-reinforcing cycle of platelet deposition. This ongoing recruitment contributes to thrombus densification, reducing permeability and resistance to therapeutic penetration. Moreover, our results indicate that new platelet populations further drive thrombus contraction, possibly through cytoskeletal rearrangements or paracrine signaling from newly adhered platelets. These findings emphasize the importance of targeting this VWF-mediated mechanism to disrupt thrombus progression and improve therapeutic efficacy. Once this active maintenance process is stopped, occlusive clots may degrade.

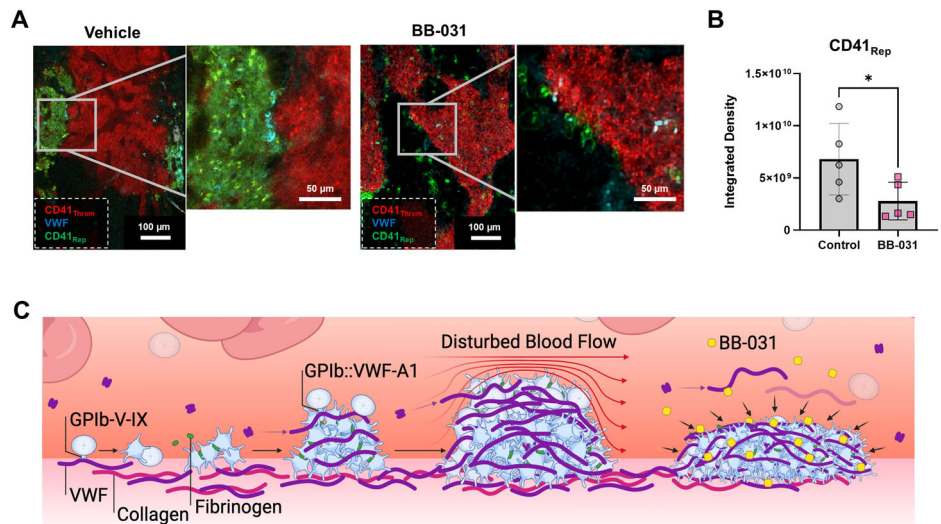
The aptamer BB-031 effectively addresses this pathological mechanism by specifically inhibiting the VWF-A1:platelet-GPIIb α interaction. Our study shows that treating thrombi with BB-031 halts the recruitment of new platelets, thereby arresting thrombus growth and facilitating channel patency. The targeted mechanism of BB-031 prevents the self-reinforcing cycle of VWF-driven platelet aggregation, providing a strategy to counteract

the densification and contraction that render thrombi resistant to fibrinolytic agents. By isolating this key pathway and by nature of the reversibility of aptamer-based technology, BB-031 thus likely offers a safer and more precise alternative to current treatments, avoiding the systemic bleeding risks associated with fibrinolytics. This mechanism may be, at least in part, akin to heparin in venous thromboembolism, where targeted inhibition of thrombin prevents clot propagation and allows endogenous fibrinolysis mechanisms to proceed³⁹. Extending this principle to high-shear arterial thrombi, BB-031 represents a paradigm shift in therapeutic strategies for arterial thrombosis.

BB-031 demonstrated superior efficacy compared to Alteplase and Tenecteplase in restoring patency in vitro. The limited effectiveness of fibrinolytics was likely attributable to the aforementioned core-shell structure of dense platelet- and VWF-rich thrombi, and ongoing thrombus contraction. These structural and compositional dynamics restrict fibrinolytic diffusion and activity within the thrombus. In contrast, BB-031 directly targets the platelet aggregation process, bypassing the structural barriers that impede fibrinolytic penetration. Furthermore, by preventing the recruitment of new platelets, BB-031 halts the densification and contraction that exacerbate thrombus resistance over time. These distinctions highlight the potential of a VWF-A1 inhibitor as a first-line therapy, especially for patients presenting with delayed treatment windows where thrombus remodeling and contraction are more pronounced. However, it is important to note that responses to BB-031 were heterogeneous. While the aptamer demonstrated overall improved efficacy, there were thrombi that were less responsive to the A1-inhibition strategy than others. This may reflect stochastic differences in thrombus composition and structure, or may reflect underlying donor genetics and biology, or other unknown mechanisms. These findings may therefore support the potential need for combination therapeutic approaches to address dynamic clot composition, perhaps at lower doses, which would potentially improve both safety and outcomes in patients with thrombotic occlusion.

Limitations of this study include a lack of endothelium in the microfluidic system. However, we put forth that the microfluidic platform utilized in this study represents an advancement in preclinical arterial thrombosis

Fig. 4 | VWF-A1 inhibition reduced new platelet deposition 3 h post-occlusion. **A** Representative median slice from z-stack confocal images (40 slices, 1.4 μm step size) (Nikon A1 218D) taken after a 2 h treatment period illustrate reocclusion of the microfluidic channel with newly reperfused platelets (CD41_{Rep} , green) following initial occlusive thrombosis ($\text{CD41}_{\text{Throm}}$, red) remodeling for 3 hr. VWF (cyan) is present at the interface of platelet populations and is diminished when treated with BB-031. **B** Quantified integrated density of newly reperfused platelets (CD41_{Rep}) is reduced in thrombi treated with BB-031 (control versus BB-031 $p = 0.0172$, $n = 5$ independent experiments). **C** Pictographic summary of VWF-A1 mediated recruitment. $*p < 0.05$. All error bars represent standard deviation. Created in BioRender. Kar, R. (2026). <https://BioRender.com/7lg93qj>.



research and can be utilized as a high-throughput discovery platform. By allowing in vitro formation of fully occlusive thrombi under arterial flow conditions, this system enables high throughput study of thrombus composition, remodeling, and therapeutic delivery in real-time. The platform intentionally isolated the contributions of blood components to thrombus dynamics, eliminating the confounding effects of endothelial-derived factors. While this lack of endothelialization was a deliberate choice to focus on platelet and VWF mechanics, future iterations could incorporate endothelial cells to further replicate in vivo conditions, particularly ischemic damage, as well as endothelial cell VWF secretion and the interplay between endogenous fibrinolysis and external therapeutic agents. Additionally, this study did not examine BB-031 efficacy in response to obesity or sex, which are known to influence VWF levels. Future studies should include these cohorts for comparison.

In this study, we utilized a microfluidic platform capable of true occlusion, thrombus remodeling post-occlusion, delivery of pharmacological agents upstream of the occlusive thrombus simulating systemic drug treatment, and quantification of the hemodynamically relevant endpoint of channel patency. This system allowed us to identify VWF-mediated platelet recruitment and reinforcement as a driver of thrombus maintenance and lytic-resistance. Inhibition of VWF-A1 with aptamer BB-031 disrupted the dynamic thrombus reinforcement process to restore blood flow. This is a promising therapeutic approach for arterial thrombosis capable of transforming management of acute thrombotic disease.

Methods

Blood sample collection

All ethical regulations relevant to human research participants were followed. Whole blood was collected from healthy volunteer donors via venipuncture after their informed consent (University of Pittsburgh Institutional Review Board (IRB) Study #22010142). A subset of these donors ($n = 10$, used for treatment comparison experiments) provided four separate donations, spaced at least seven days apart, to allow paired comparisons across treatment groups. Samples were drawn into sodium heparin tubes (BD Vacutainer 367874) for use in microfluidic assays and into EDTA tubes (BD Vacutainer 367841) for complete blood counts (CBCs), which were performed using a hematology analyzer (Advia 2120i; Siemens, Munich, Germany). Whole blood was also collected from patients diagnosed with AIS at presentation at The Ohio State Wexner Medical Center before pharmacological treatment or mechanical thrombectomy was performed by the Department of Neurosurgery biorepository (NSBR) staff. Eligible participants or their legally authorized representative (LAR) was approached regarding study participation. The approved study informed consent form

(ICF), or translated approved consent form, were provided to the potential participant or their LAR by an appropriately delegated member of the research team. In the case of an emergent procedure, potential participants or LARs were consented after samples were collected during their hospital stay or within 30 days of the collection of specimens. Patient information was removed from all samples and a unique patient code was assigned. NSBR staff obtained consent and whole blood was collected in tubes containing lithium heparin before being transferred within 10 min of draw to laboratory staff. All experiments on healthy cohort or AIS patient samples began within 1 h of collection.

Microfluidic device preparation

Custom brass molds for microfluidic devices with 8 parallel wye channels were fabricated using microcomputer numerical control (CNC) milling at the University of Pittsburgh Swanson Center for Production Innovation. The stenotic wye channel design featured a width of 480 μm , stenosis height of 80 μm , and a height of 180 μm in non-stenotic regions. Polydimethylsiloxane (PDMS) was mixed at a 10:1 elastomer-to-curing agent ratio, poured into the molds, and cured. Inlet and outlet ports were created using 1.5 mm biopsy punches. Cured PDMS molds were bonded to glass slides using a Harrick Plasma cleaner (Harrick Plasma, Ithaca, NY, USA). Channels were sterilized with 70% ethanol. The stenotic region was coated overnight (~ 18 h) with a 5:1 (v/v) 0.9% sodium chloride to collagen solution (equine fibrillar type I, Chrono-Log, Havertown, PA, USA). Before experiments, channels were rinsed with phosphate-buffered saline (PBS, pH 7.4). The design, mold fabrication, PDMS cast fabrication, and channel fabrication were adapted from a previously established stenotic microfluidic model without a wye^{31,40–42}.

Microfluidic system for occlusive thrombus formation

Thrombi were generated by perfusing anticoagulated whole blood through the microfluidic device at a constant pressure head of 20 mmHg (Fig. 1A–C). Microfluidic mass outflow was measured to calibrate flow rate (Supplementary Fig. 1), and computational fluid dynamics simulations (Supplementary Fig. 2) were conducted to model initial shear rate using Ansys 2025 R2 (25.2) and the following parameters: mesh size 1e-6 m (1 micron), no-slip conditions at wall, inlet pressure of 20 mmHg, outlet open and 0 mmHg gauge pressure, and convergence criteria for residuals of $\leq 1\text{e-}6$. Simulations generated a pathophysiologically-relevant arterial shear rate of 15,357 s^{-1} within the stenosis region.

Whole blood was pre-incubated with human Fc receptor block (Biolegend 422302; 1:600) and anti-CD41:JF646 (Novus Biologicals NB100-2614JF646; 1:1200) unless otherwise noted for 30 min before

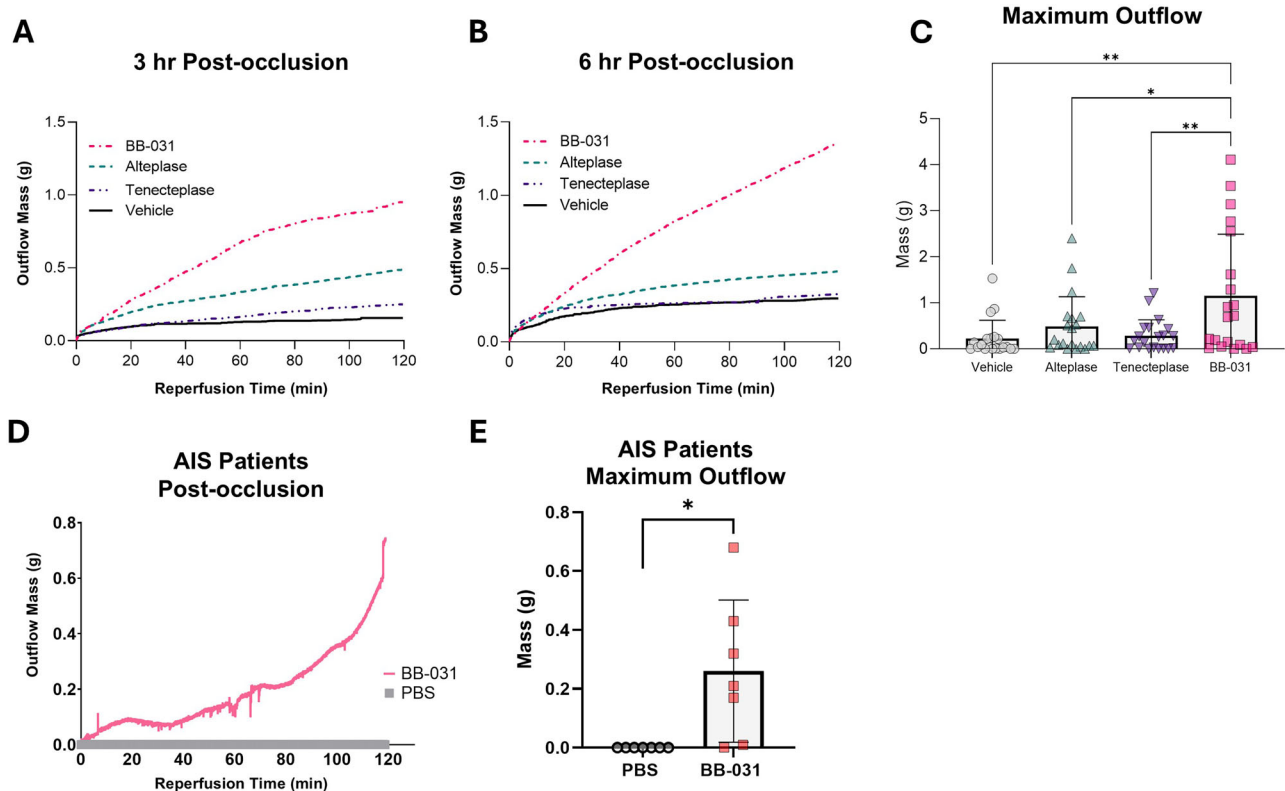


Fig. 5 | VWF-A1 inhibitor (BB-031) improves patency in healthy and AIS patient samples. **A** 3 h post-occlusion, inhibition of VWF-A1 over a 2 h treatment period (via BB-031) increased microfluidic outflow mass. **B** 6 h post-occlusion, BB-031 demonstrated similar improvement in patency. **C** Combined final outflow mass from 3 and 6 h post-occlusion, demonstrates significant improvement in patency with BB031 compared to vehicle ($p = 0.0021$), Alteplase ($p = 0.0437$), and Tenecteplase ($p = 0.0047$) from $n = 20$

independent experiments. **D** AIS patient samples show similar patency improvements post-occlusion with BB-031 treatment compared to vehicle. **E** Maximum mass outflow during drug treatment period demonstrates significantly higher mass outflow in BB-031 treated thrombi compared to vehicle treated thrombi ($p = 0.0146$, $n = 7$ independent experiments) for AIS patient samples. $*p < 0.05$. $**p < 0.01$. All error bars represent standard deviation.

perfusion of the blood sample through the primary outlet with the secondary outlet closed. Occlusive thrombus formation in our system was defined as a clot able to cause cessation of flow against 20 mmHg pressure head for 3 min. Thrombus formation resulting in cessation of flow was noted visually in real-time using fluorescence microscopy (Zeiss, Axio Observer 7) (Supplementary Movie 1: “Thrombus formation.avi”) and confirmed in real-time by monitoring downstream mass outflow for 3 min using an analytical scale (OHAUS 30253019) and LabView software. Following occlusion, device inlet and outlets were closed. Representative z-stack images of thrombi 3 h post-occlusion were captured using confocal microscopy (Olympus FV4000) with blood pre-incubated with Fc receptor block (1:600), anti-CD41:JF646 (1:1200), 488 labeled fibrinogen (Invitrogen F13191; 10 $\mu\text{g/mL}$), and anti-VWF:550 (Novus Biologicals NBP2-54379R; 1:900).

Structural thrombus comparisons ($n = 3$) 0 h and 3 h post-occlusion using whole blood collected in alternative whole blood anticoagulants (citrate, citrate with recalcification (10 mM CaCl_2 immediately prior to perfusion), and heparin) was also conducted. Blood was pre-incubated with Fc receptor block (1:600), anti-CD41:JF646 (1:1200), 488 labeled fibrinogen (50 $\mu\text{g/mL}$), and anti-VWF:550 (1:600). Representative confocal images (Zeiss LSM900) are shown in Supplementary Fig. 3 where citrate anticoagulant on occasion depicts robust fibrin formation, however, with recalcification shows the least fibrin formation potentially due to consumption upstream of the stenotic region.

Thrombus porosity analysis post-occlusion

Thrombi ($n = 60$) were generated as described above and imaged using real-time fluorescence microscopy (Zeiss, Axio Observer 7). Post-occlusion,

device inlets and outlets were closed, and thrombi were left undisturbed under static pressure conditions for 0, 3, or 6 h ($n = 20$ per group) to mimic temporal clot remodeling. Pore areas were quantified using ImageJ software via binary thresholding and a gating strategy that identified central pores with areas larger than 28 μm^2 . Images taken at thrombus occlusion and 0, 3, and 6 h post-occlusion following device inlet and outlet closure were quantified.

Thrombus composition retraction and remodeling

For spatial remodeling analysis, whole blood ($n = 3$) was incubated with human Fc receptor block (1:600), anti-CD41:JF646 (1:1200), 488 labeled fibrinogen (50 $\mu\text{g/mL}$), and anti-VWF:550 (1:600) and perfused through the microfluidic device as described above. Post-occlusion, device inlets and outlets were closed, and thrombi were left under static pressure conditions for 0, 3, or 6 h. Z-stack images of thrombi were captured using confocal microscopy (Olympus FV4000).

Visualization of VWF-platelet interactions in thrombus development

For studies of VWF-platelet interactions in the developing thrombus ($n = 3$), whole blood was incubated with human Fc receptor block (1:600), anti-CD41:JF646 (1:600), and anti-VWF:550 (1:900) and perfused through the microfluidic device. Once platelet adhesion was confirmed microscopically, the channel was flushed with 10% neutral buffered formalin (NBF) followed by PBS twice. Platelets in this original thrombus are labeled CD41^{Throm}. For subsequent platelet incorporation experiments, a second aliquot of whole blood pre-incubated with anti-CD41:AF488 (Novus Biologicals NB100-2614AF488; 1:600) was perfused through the

same device under identical conditions prior to NBF and PBS as described above. These “reperfused” platelets are labeled CD41_{Rep}. Fixed thrombi were imaged using confocal microscopy (Olympus FV4000 and Zeiss LSM900).

Thrombus retraction with reperfused platelets

For retraction experiments with reperfused platelets ($n = 3$), whole blood was incubated with human Fc receptor block (1:600) and anti-CD41:JF646 (1:600) and perfused through the microfluidic device to occlusion as described above. Post-occlusion, thrombi were aged under static pressure for 6 h. Platelets in this original thrombus are labeled CD41_{Throm}. Following a PBS flush, an aliquot from the original whole blood sample pre-incubated with human Fc receptor block (1:600) and anti-CD41:AF488 (1:600) was flowed through the secondary outlet before reperfusion over the stenotic region was established at a pressure of 2.7 mmHg, generating a shear rate of $\sim 1,000 \text{ s}^{-1}$ for 2 h. Repperfused platelets are labeled CD41_{Rep}. Fluorescence microscopy (Zeiss, Axio Observer 7) captured images at the following time points: during real-time thrombus formation, 0, 3, and 6 h post-occlusion, and after 2 h reperfusion. Platelet surface area was quantified using ImageJ and normalized to 0–100%.

VWF-platelet interactions in remodeled thrombi treated with VWF-A1 inhibitor

For experiments investigating VWF-mediated interactions in remodeled thrombi ($n = 5$), whole blood was incubated with human Fc receptor block (1:600), anti-CD41:JF646 (1:600), and anti-VWF:550 (1:900) and perfused through the microfluidic device as described above (CD41_{Throm}). Post-occlusion (3 h), aliquots from the original whole blood sample were incubated with human Fc receptor block (1:600) and anti-CD41:AF488 (1:600) and mixed with either vehicle (0.9% NaCl) or VWF-A1 inhibitor BB-031 (1692 nM). After a PBS flush through the secondary outlet, treated blood was perfused through the secondary outlet before reperfusion over the stenotic region was established at a pressure head of 2.7 mmHg, generating a shear rate of $\sim 1000 \text{ s}^{-1}$ (CD41_{Rep}). After 2 h reperfusion, the primary outlet was closed, and the secondary outlet was opened to perfuse NBF until detectable at the channel bifurcation. The secondary outlet was then closed, and the primary outlet reopened, with NBF delivery maintained at 2.7 mmHg for 30 min. The thrombus was subsequently rinsed twice with PBS under the same pressure conditions. Z-stack images of thrombi were captured using confocal microscopy (Nikon A1 218D), and integrated fluorescence density of reperfused platelets was calculated using Python 3.13.0.

Comparative drug delivery and microfluidic patency measurements in healthy volunteers and AIS patient samples

For drug comparison experiments in healthy samples ($n = 10$, paired for treatment comparisons), thrombi were formed as described above (CD41_{Throm}). Post-occlusion (0, 3, or 6 hr), aliquots from the original whole blood sample were incubated with human Fc receptor block (1:600) and anti-CD41:JF549 (1:1200) (CD41_{Rep}) and mixed with vehicle (0.9% NaCl), BB-031 (1692 nM), Alteplase (0.7 nM), or Tenecteplase (0.7 nM). Following a PBS rinse through the secondary outlet, the aliquot of blood sample mixed with treatment was perfused through the secondary outlet before reperfusion over the stenotic region was established at a pressure head of 2.7 mmHg ($\sim 1000 \text{ s}^{-1}$) for 2 h. Flow out of the microfluidic device (indicating recanalization and patency) was monitored by measuring mass outflow and real-time fluorescent images captured (Zeiss, Axio Observer 7).

For drug delivery experiments with AIS patient samples ($n = 8/\text{group}$), thrombi were formed and reperfused as described above. Thrombi were immediately reperfused with blood treated with either vehicle (0.9% NaCl) or BB-031 (1692 nM; approximating 1 mg/kg). Flow out of the microfluidic device (indicating recanalization and patency) was monitored for 2 h by measuring mass outflow.

Statistics and reproducibility

A mixed-effects analysis was performed with Tukey multiple comparisons at a 95% confidence interval for thrombus pore area. A paired two-tailed t -test was performed to compare normalized clot surface area in contraction experiments and integrated density of reperfused platelets in confocal experiments between BB-031 and vehicle conditions. Outflow mass curves from healthy samples were corrected for evaporation. For maximum mass outflow an ordinary one-way analysis of variance was performed with Tukey multiple comparisons at a 95% confidence interval. For AIS patient mass curves, outliers were removed via the ROUT method and mean curves corrected for evaporation. For AIS patient maximum outflow, outliers were removed via the ROUT method and an unpaired two-tailed t -test was performed.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data underlying the graphs presented in the main figures can be found in Supplementary Data. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Code availability

Code used to facilitate image processing and data handling, while not essential to repeat the published results, are available from the corresponding author on reasonable request.

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Author contributions

E.P.M. and R.K. contributed to experimental design, performed the studies, analyzed data, and led manuscript writing and preparation. R.M.G.R. designed and refined the model, contributed to experimental design, performed the studies, analyzed data, contributed to the initial manuscript draft, and reviewed and edited the manuscript. B.H., D.W., and R.S. performed the studies, analyzed data, and reviewed and edited the manuscript. S.M.N. provided internal funding at OSU, reviewed analysis, and reviewed and edited the manuscript. S.M.S. provided internal funding at Pitt, led research design, reviewed analysis, and reviewed and edited the manuscript.

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Competing interests

S.M.N. is founder and CMO of Basking Biosciences. Basking Biosciences provided a portion of funding for this study to S.M.S. at University of Pittsburgh, and also provided BB-031 material for the study. All other authors declare no competing interests.

Additional information

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